

# Bala Arun Pasala

Northampton, MA | +1 4133818673 | [balaarunpasala.dev@gmail.com](mailto:balaarunpasala.dev@gmail.com) | [LinkedIn](#) | [Github](#)

## PROFESSIONAL SUMMARY

Software Engineer with 3.5+ years of experience building scalable backend systems, microservices, and RESTful APIs using Java, Spring Boot, Node.js, and Express. Experienced in event-driven architectures, real-time processing, database optimization, and cloud-based deployments. Skilled in React, MongoDB, SQL, Docker, CI/CD, and API integrations, including healthcare interoperability and AI-driven workflows. Proven track record of improving performance, automating complex processes, and delivering reliable, user-focused solutions in Agile environments across SaaS, healthcare, fleet operations, and data-intensive platforms globally.

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, SQL, Python

**Frameworks:** Spring Boot, Node.js, Express.js, React.js, Tailwind CSS

**Backend:** REST APIs, Microservices, WebSockets, Distributed Systems

**Databases:** MongoDB, PostgreSQL, MySQL

**Cloud and DevOps:** AWS, Azure, Docker, Jenkins, GitHub Actions, CI/CD

**Integration:** HL7 v2, FHIR, JSON, XML, Third-Party APIs

**Security:** OAuth 2.0, JWT, Authentication, Authorization

**Testing and Tools:** Postman, JUnit, Git, GitHub, GitLab, Jira

**AI and Analytics:** AI Integration, Machine Learning Services, Route Optimization, Predictive Maintenance, Computer Vision

## PROFESSIONAL EXPERIENCES

### Software Engineer Associate

ScalaCode | USA | Jun 2025 – Present

- Developed and maintained scalable microservices and RESTful APIs using Node.js, Express, and Spring Boot to power real-time fleet operations, AI-driven workflows, and core SaaS platform functionality.
- Engineered real-time and event-driven backend systems using WebSockets and asynchronous messaging to process 10,000+ events per day, enable low-latency live tracking, and reduce dashboard update delays by 35%.
- Designed and optimized data models and query patterns across MongoDB, SQL, and time-series workloads, improving query performance by 40% and supporting analytics across 1M+ records with faster data retrieval.
- Integrated AI and machine learning services, including route optimization, predictive maintenance, and computer vision-based blueprint analysis, transforming raw outputs into structured insights that improved automation and decision-making.
- Built end-to-end file ingestion and processing pipelines for large blueprint and operational datasets, supporting files up to 100 MB, reducing manual processing effort by 50%, and improving estimation turnaround time by 40%.
- Developed responsive frontend components and dashboards using React, React Hooks, and Tailwind CSS, simplifying complex workflows and improving usability across tracking and estimation platforms.
- Contributed to cloud deployment, CI/CD automation, monitoring, and Agile software delivery using AWS or Azure, Docker, GitHub Actions or Jenkins, Git, Jira, Postman, and observability tools to improve release reliability and production readiness.

### Software Engineer

Verily | India | Jan 2022 – Jul 2024

- Build and maintained scalable backend applications in Java and Spring Boot, building RESTful APIs and distributed services to support healthcare data interoperability, claims workflows, and clinical operations.
- Designed and integrated APIs for patient data exchange, eligibility verification, and third-party connectivity, boosting processing efficiency by 60% while reducing transaction failures through resilient retry handling.
- Engineered data transformation and workflow automation pipelines for HL7 v2, FHIR, JSON, and XML data, reducing manual processing effort by 40% and improving consistency across EHR, laboratory, and claims ecosystems.
- Designed, optimized, and managed relational and NoSQL database schemas across PostgreSQL, MySQL, and MongoDB, enabling efficient storage and retrieval of patient records, clinical observations, claims data, and validation results.
- Implemented secure authentication and authorization frameworks using OAuth 2.0 and JWT, strengthening access controls for sensitive healthcare data and supporting security and compliance requirements.
- Performed API testing, debugging, and production issue resolution using Postman, JUnit, and log analysis, improving system reliability by 35% and reducing integration issues by 40% across internal and external applications.
- Collaborated in Agile development environments using Git, GitHub, GitLab, Jira, Docker, Jenkins, and GitHub Actions to deliver production-ready features, participate in code reviews, and streamline CI/CD workflows.

## EDUCATION

Master of Science in Computer Science | May 2026

Western New England University | Springfield, MA

Bachelors in Computer Science | Mar 2022

St. Joseph's Autonomous | Museum Road, Bengaluru, India

## PROJECTS

### AI-Based Student Performance Prediction System

- Implemented a machine learning model using Python (Scikit-learn) to predict student performance based on academic and behavioral data, enabling early identification of at-risk students.
- Built a full-stack web application using React.js and Node.js with MySQL featuring interactive dashboards and data-driven insights for educators.

### Circle Save - Secure Savings Circle Platform (Capstone Project)

- Designed and developed a full-stack fintech platform for community savings groups (ROSCA), with a strong focus on scalability, security, transparency, and auditability.
- Built secure backend systems including JWT authentication, OTP email verification, role-based access control, and REST APIs for managing users, circles, memberships, and transactions.
- Implemented end-to-end circle workflows such as creation, join approvals, contribution tracking, notifications, and backend validation to reduce risks like duplicate payments and unauthorized access.

## CERTIFICATION

- Java Full stack training (JSPIDERS)